CS 210

Name_____ RedID __ __ __ __ __ __ __ __ __

Program 3 Analysis

https://github.com/kadenhill693/Program-2

1. Briefly define a *queue* data structure.
   A queue is on a FIFO order like a line taking customers. New things go to the back while the oldest get put out

   What evidence did you find that a *queue* container was operating in the load balancer?
   The requests were taken care of in the order that they came in. The spots given went from 1-5 at first then gradually increased. Regardless of how long they were, they were sorted in the way that they came in and left like that

2. Briefly define a *stack* data structure.
   A stack is LIFO structured. You add to the top and you take from the top, so the most recent thing gets pulled out

   What evidence did you find that a *stack* container was operating in the load balancer?
   The newest requests got taken care of first. The spots given started from 10-6 instead of 1-5. Newer requests got picked up quickly and taken care of and the order of things was all over the place other than the fact that the order leaned towards bigger numbers

3. Briefly define a *priority queue* data structure.
   A priority queue is the same as a queue but everything has a certain priority and the highest priority goes first. It's like vip treatment at any public place

   What evidence did you find that a *Max priority queue based on process time* was operating in the load balancer?

   The longest requests got served first and a lot of what was finished was very long. The processing time took much longer compared to the others and the waiting line got really long

   What evidence did you find that a *Min priority queue based on process time* was operating in the load balancer?

   The shortest requests were served first and the requests that were finished were very short. The average time to process was shorter than the others and the longest requests were pushed behind all the shortest ones

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Evidence can be a description of the output specific to the data structure and/or an annotated image.

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What evidence did you find that a *Max priority queue based on process ID* was operating in the load balancer?

The newest requests were served first and higher numbers in order quickly got served as they were requested. The processing time was very similar to the stack as the stack and the max priority queue are pretty similar

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Evidence can be a description of the output specific to the data structure and/or an annotated image.